

Transforming Quadratic Functions to Vertex Form

Jonathan R. Bacolod

The **vertex form** of a quadratic function is

$$y = a(x - h)^2 + k$$

The Vertex

The **vertex** is the point (h, k) .

$$y = a(x - h)^2 + k$$

For example, in $y = 2(x - 3)^2 + 4$, the vertex is $(3, 4)$.

The Role of a

The value of a determines the **opening direction** and the **width** of the parabola.

- ▶ If $a > 0$, the parabola opens **upward**.
- ▶ If $a < 0$, the parabola opens **downward**.
- ▶ If $|a| > 1$, the parabola is **narrower**.
- ▶ If $0 < |a| < 1$, the parabola is **wider**.

Already in Vertex-Like Form

If the equation is already in the form

$$y = a(x - b)^2$$

then simply write it as

$$y = a(x - b)^2 + 0$$

Forms $y = ax^2$ or $y = ax^2 + b$

For equations of the form $y = ax^2$ or $y = ax^2 + b$, write them as

$$y = a(x - 0)^2 + k$$

Standard Form

For equations of the form

$$y = ax^2 + bx + c$$

use the method of **completing the square**.

Completing the Square

1. Divide the coefficient of x by 2.
2. Square the result.
3. Add and subtract this value inside the expression.
4. Rewrite the perfect square trinomial as $(x - h)^2$.

Examples

Example 1

Transform $y = x^2 + 6x + 7$ into vertex form.

$$y = (x^2 + 6x) + 7$$

Half of 6 is 3, and $3^2 = 9$.

$$y = (x^2 + 6x + 9 - 9) + 7$$

$$y = (x + 3)^2 - 2$$

Example 2

Transform $y = 2x^2 + 8x + 3$ into vertex form.

$$y = 2(x^2 + 4x) + 3$$

Half of 4 is 2, and $2^2 = 4$.

$$y = 2(x^2 + 4x + 4 - 4) + 3$$

$$y = 2(x + 2)^2 - 5$$

Example 3

Transform $y = -x^2 + 4x + 1$ into vertex form.

$$y = -(x^2 - 4x) + 1$$

Half of -4 is -2 , and $(-2)^2 = 4$.

$$y = -(x^2 - 4x + 4 - 4) + 1$$

$$y = -(x - 2)^2 + 5$$

Practice Exercises

Practice 1

Transform into the form $y = a(x - h)^2 + k$.

$$y = 2x^2$$

Practice 2

Transform into the form $y = a(x - h)^2 + k$.

$$y = -5x^2 + 4$$

Practice 3

Transform into the form $y = a(x - h)^2 + k$.

$$y = 3(x - 2)^2$$

Practice 4

Transform into the form $y = a(x - h)^2 + k$.

$$y = -4(x + 1)^2$$

Practice 5

Transform into the form $y = a(x - h)^2 + k$.

$$y = x^2 + 8x + 10$$

Practice 6

Transform into the form $y = a(x - h)^2 + k$.

$$y = x^2 - 12x + 20$$

Practice 7

Transform into the form $y = a(x - h)^2 + k$.

$$y = 2x^2 + 12x + 5$$

Practice 8

Transform into the form $y = a(x - h)^2 + k$.

$$y = 3x^2 - 18x + 6$$

Practice 9

Transform into the form $y = a(x - h)^2 + k$.

$$y = -x^2 + 4x - 7$$

Practice 10

Transform into the form $y = a(x - h)^2 + k$.

$$y = -2x^2 - 8x + 3$$

Activities

Activity 1

Transform into the form $y = a(x - h)^2 + k$.

$$y = 7x^2$$

Activity 2

Transform into the form $y = a(x - h)^2 + k$.

$$y = -3x^2 - 6$$

Activity 3

Transform into the form $y = a(x - h)^2 + k$.

$$y = 5(x + 4)^2$$

Activity 4

Transform into the form $y = a(x - h)^2 + k$.

$$y = -2(x - 6)^2$$

Activity 5

Transform into the form $y = a(x - h)^2 + k$.

$$y = x^2 + 10x + 1$$

Activity 6

Transform into the form $y = a(x - h)^2 + k$.

$$y = x^2 - 4x + 9$$

Activity 7

Transform into the form $y = a(x - h)^2 + k$.

$$y = 2x^2 - 16x + 7$$

Activity 8

Transform into the form $y = a(x - h)^2 + k$.

$$y = 4x^2 + 24x - 8$$

Activity 9

Transform into the form $y = a(x - h)^2 + k$.

$$y = -x^2 - 10x + 6$$

Activity 10

Transform into the form $y = a(x - h)^2 + k$.

$$y = -3x^2 + 18x - 5$$

Thank you!